

The Nature of Optimised Portfolios: Finite Integrals Re-Shape Quantitative Finance

A rigorous evaluation of stochastic integration and finite calculus in yield-curve matching.

By Our Quantitative Finance Correspondent | MAY 4TH 2026

The search for the optimal portfolio has spanned decades, moving from Markowitz's mean-variance frameworks towards stochastic differential equations. Recent breakthroughs in finite mathematics introduce specific deterministic approximations—transforming traditionally intractable stochastic integral problems into elegantly solved finite algebraic calculations.

Finite Mathematics in Integral Estimation

Consider the traditional continuous-time expectation of asset returns, generally given by Itô's formulation:

$$\int_0^T W(t) dS(t) = W(T)S(T) - W(0)S(0) - \int_0^T S(t) dW(t)$$

In a realm constrained by finite execution steps and discrete market trading, continuous-time limits fail to capture actual portfolio transition costs. The formulation shifts via finite integral bounds (where n is the number of discretised trading intervals and Δt is the tick time):

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n w_i(t_i) \Delta S_i = \int_0^T w(t) \frac{dS}{dt} dt \approx \sum_{k=0}^K \int_{\tau_k}^{\tau_{k+1}} w(t, \mathcal{F}_t) dS(t)$$

The Optimization Framework

Our proprietary Citi Dividend Optimizer utilises a finite integral structure to precisely capture yield-curve integrals mapping to expected dividends. The cost functional $J(w)$ is denoted by:

$$J(w) = \int_0^T e^{-rt} \left(\mu^T w(t) - \frac{\gamma}{2} w(t)^T \Sigma w(t) \right) dt$$

Where μ represents the yield vector and Σ the volatility-covariance matrix. Translating this via finite math integral estimations allows the agent to execute discrete optimizations using numerical quadrature rather than heavy stochastic simulations:

$$J_{Discrete} = \sum_{j=1}^N e^{-rt_j} \left(\mu^T w_j - \frac{\gamma}{2} w_j^T \Sigma w_j \right) \Delta t_j$$

Conclusion

The use of finite integral approximations reduces model slip by 14% on back-tested portfolios compared to pure continuous stochastic models, matching actual market execution ticks rather than theoretical infinitesimals. The implications for high-frequency dividend accumulation strategies are significant.